

17. Ans: C

The electric force on the positive charge is pointing in the direction of the E-field. So the external force exerted to move the charge is pointing in the opposite direction to the E-field.

$$\begin{aligned}\text{Work done by external force, } W &= F s \cos \theta = qEs \cos 0^\circ \\ &= (2.6 \times 10^{-8})(3.0 \times 10^5)(4.0 \times 10^{-3}) \\ &= + 3.1 \times 10^{-5} \text{ J}\end{aligned}$$

18. Ans: A